

Unit 7 Guided Notes — Data Analysis & Statistics

Scatterplots, correlation, the line of best fit, and two-way tables.

Standards: NC.8.SP.1 · NC.8.SP.2 · NC.8.SP.3 · NC.8.SP.4

 **Blank Worksheet (PDF)**

 **Answer Key (PDF)**

 **Official PDF Packet**

 **EC/Modified Notes (PDF)**

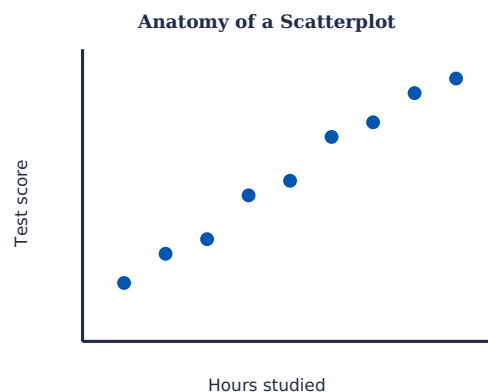
Interactive guided notes for Unit 7 — scatterplots, correlation, the line of best fit, and two-way tables.
Fill in the blanks to learn, then click to check your thinking.

1 Scatterplots & Bivariate Data

✦ **NC.8.SP.1:** Construct and interpret scatter plots for bivariate measurement data.

Bivariate data has TWO values for each item — like a person's height AND weight, or hours studied AND test score. We use a scatterplot to look for patterns between the two variables.

Term	Meaning	Example
Univariate data	ONE variable per data point	a list of test scores
Bivariate data	TWO variables per data point	hours studied AND test score



A scatterplot has: an x-axis for the **independent (input)** variable, a y-axis for the **dependent (output)** variable. Each dot represents **one** data point. Both axes need a **label** and a **scale**.

Try It

Suppose you collect data on 10 students: how many hours of sleep they got and their test score the next day.

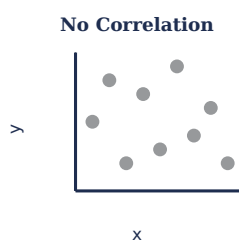
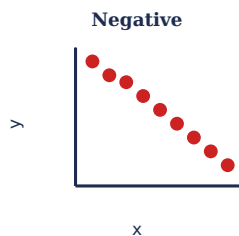
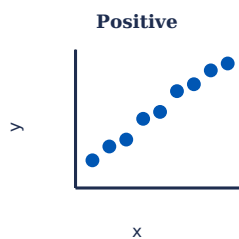
- Independent variable (x-axis)? **Hours of sleep**
- Dependent variable (y-axis)? **Test score**
- Total number of dots on the scatterplot: **10**

💡 **Why scatterplots?** Bar graphs show frequencies of categories. Scatterplots show relationships between two numerical quantities. If you're asking "does X affect Y?", you want a scatterplot.

2 Correlation

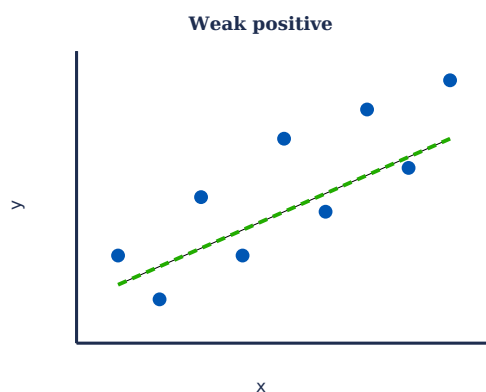
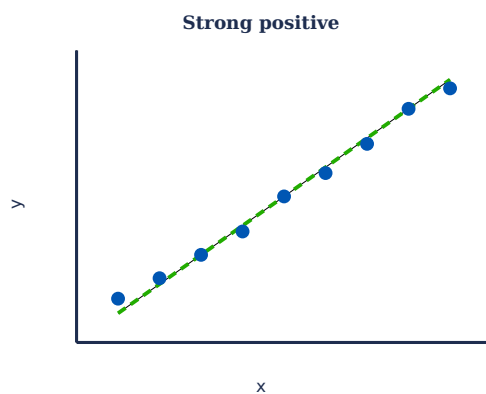
✦ **NC.8.SP.1:** Describe patterns such as clustering, outliers, positive or negative association, linear association, and nonlinear association.

Correlation describes how two variables tend to change together — **positive**, **negative**, or **none**. Plus two strengths: **strong** or **weak**.



Positive correlation: as x increases, y also **increases** . Negative correlation: as x increases, y **decreases** . No correlation: the dots are scattered with **no** clear pattern.

Strength of Correlation



Strong = dots tightly hug the trend line. Weak = dots are spread out around the trend.

Try It — Predict the correlation type

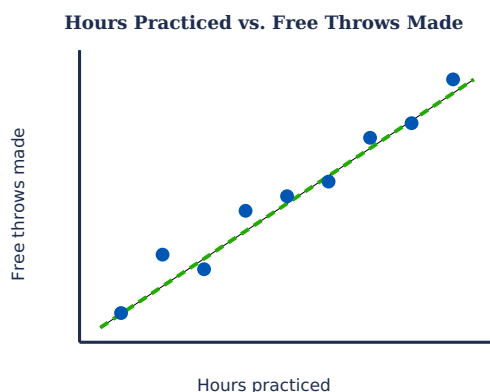
- Hours studied vs. test score → **Positive**
- Number of absences vs. final grade → **Negative**
- Shoe size vs. IQ → **None**
- Outdoor temperature vs. ice cream sales → **Positive**
- Age of a car vs. value of the car → **Negative**

⚠ **CORRELATION $\neq$ CAUSATION:** Just because two things are correlated doesn't mean one causes the other! Ice cream sales and shark attacks are both higher in summer — but ice cream doesn't cause sharks. Both are caused by hot weather.

3 Line of Best Fit

✦ **NC.8.SP.2 / NC.8.SP.3:** Informally fit a straight line for scatter plots that suggest a linear association; use the equation to solve problems and make predictions.

When the dots suggest a linear pattern, we can draw a single straight line through the "middle" of the cloud of dots. This line represents the overall trend and helps us make predictions.



Notice the line: passes through the "middle" of the dots · has roughly the same number of dots above and below it · follows the overall positive trend · doesn't have to touch any specific dot.

Drawing a Line of Best Fit (Informally)

1. Look at the overall direction of the dots (positive or negative).
2. Use a ruler — put the line so dots are roughly balanced above and below.
3. Don't connect the dots! It's a smooth straight line, not a zigzag.
4. Don't be tricked by outliers — ignore dots far from the rest.

Try It — $y = 2x + 1$ (x = hours studied, y = test score)

- The slope means: each extra hour of study adds about **2** points.
- The y-intercept means: with 0 hours of study, the predicted score is **1**.
- If a student studies 4 hours, predicted score = $2(4) + 1 =$ **9**.
- If a student studies 8 hours, predicted score = **17**.

⚠ **Predictions are approximate.** The line is a model, not exact. A student who studies 4 hours might get 7 or 11 — the line predicts about 9 on average.

4 Two-Way Tables

✦ **NC.8.SP.4:** Understand patterns of association between two variables by displaying frequencies and relative frequencies in a two-way table.

Two-way tables organize categorical bivariate data — when both variables are categories instead of numbers. Example: grade level and lunch preference.

Reading a Two-Way Table — 120 students surveyed about pizza

	Like Pizza	Don't Like Pizza	Total
6th Grade	32	8	40
7th Grade	28	12	40
8th Grade	24	16	40
Total	84	36	120

- Total students surveyed: **120**
- Number of 6th graders who like pizza: **32**
- Total 7th graders surveyed: **40**
- Total students who don't like pizza: **36**

Relative Frequency

Relative frequency = the count divided by a total.

- Fraction of all 120 students who like pizza: $84 \div 120 =$ **0.7** or **70%**
- Fraction of 6th graders who like pizza: $32 \div 40 =$ **0.8** or **80%**
- Fraction of 8th graders who like pizza: $24 \div 40 =$ **0.6** or **60%**

💡 **Looking for an association:** 80% of 6th graders like pizza but only 60% of 8th graders do. There's an association between grade level and pizza preference — they're related!

Try It — 100 students surveyed about pets

	Has pet	No pet	Total
Boys	20	30	50
Girls	35	15	50
Total	55	45	100

- Fraction of boys who have a pet: $20 \div 50 =$ **0.4** or **40%**
- Fraction of girls who have a pet: $35 \div 50 =$ **0.7** or **70%**
- Is there an association between gender and pet ownership?

Yes – girls are much more likely to have one

Error Analysis

SPOT THE ERROR "Ice cream sales and drowning rates have a positive correlation, so eating ice cream causes drowning."

Correlation does NOT mean causation! Both increase in summer because of a lurking variable — hot weather. Always look for a third variable that could explain the relationship.

SPOT THE ERROR "The line of best fit must pass through every data point."

The line of best fit shows the general trend — it should have roughly equal points above and below, but does NOT need to pass through any specific point.

SPOT THE ERROR "More 8th graders like basketball (35) than 7th graders (30), so 8th graders are more into basketball." (8th grade has 80 students total vs 70 in 7th grade!)

Use relative frequency to compare fairly: 7th = $30/70 \approx 42.9\%$ vs 8th = $35/80 \approx 43.75\%$. They're nearly the same! When comparing groups of different sizes, ALWAYS use percentages, not raw counts.

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